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Humans power technology. Simple truth. Machines have become such an integral part of our lives that we subconsciously accept them as beings, which is probably why we get so annoyed when they act like idiots. While we liberally abuse Windows, Google Image Search, voice recognition software and countless other software and services, we forget that old gem—GIGO (garbage in, garbage out). Write bad code, get bad software. Muck up a search query, get irrelevant search results.

“Help me help you,” says the salesman to his customer, says the doctor to his patient, says the PC to its user. Funny how that last bit never occurs to us. Your computer won’t do your taxes of its own volition, and the Web won’t give you the right answers unless you ask the right questions. Technology won’t serve us better till we help it do so.

The Old-fashioned Way

The best way to get help, of course, is to ask for it. One of the most challenging tasks for a computer is labelling images so they’ll turn up in relevant search queries. A program can’t know that DSC00023.jpg is a picture of a pigeon

Then again, not many people are always willing to help...

Getting Sneaky, Part I

Image labelling isn’t the only way you can do your bit—in fact, there’s another thing you’re helping with already. When you register for any community site (forums, blogs and more), you’re asked to “prove that you’re human” by typing the word you see in a verification image. Instead of the computer using complex AI-like algorithms to detect spam-bot behaviour, it just asks you to use your human intelligence, stumping the bots. It’s called CAPTCHA, and now, Carnegie Mellon’s Luis von Ahn is taking this one step ahead with reCAPTCHA.

Right now, the Internet Archive (www.archive.org), among other groups, is digitising classic books to create an online repository of books that were written before the computers came. The physical books are being scanned and given to OCR (Optical Character Recognition) programs—all very well, but there are some snags. OCR software isn’t perfect, so a lot of words come out misread (luckily, most programs tell you that there may have been an error). The only way to get the right word is to have it read by a human, and with the volumes of text we’re talking about, a small group isn’t

Giving Back To

In the quest to make machines work better, we’ve finally hit upon the secret ingredient—us!

in a banyan tree, and will never be able to tell—unless it’s truly intelligent, and we all now know enough not to hold our breath for AI. Humans, on the other hand, can immediately come up with a whole bunch of labels for any image. But the show must go on, so somewhere in the world, someone is tagging these images so you can see it when you type in “pigeon tree” in an image search. It’s still an incredible task for a small group of people, but what if all of us got involved?

The concept is simple. Sign up at Google Image Labeler (<http://images.google.com/imagelabeler/>) or the ESP Game (<http://www.espgame.org/>) and play the image labelling game. You get paired with a random partner from a random corner of the world, and you’re both shown the same image. Both of you spew keywords to describe the image, and when your keywords match, you both get points. You get your few minutes of entertainment (not to mention the pride of earning points for it), Google’s Image Search algorithm gets away with being just a bit of code, and a satisfied user somewhere gets the search result he wanted. Everyone’s happy.

feasible. So von Ahn decided to take advantage of the fact that people solve 60 million CAPTCHAs a day—if each CAPTCHA was replaced with a word that the OCR software was having a problem with, then people would not only be proving their humanity to the site, they’d also be helping digitise the books without errors! Every unrecognised word is paired with another word that the program knows the answer for; if you get one right, it assumes that you got the other one right too. It doesn’t stop there, though—now that you know how it works, you’ll likely try to see if you can get away with typing gibberish for the second word (you know you want to). To counter this, the program accepts the answer for the unknown word only after a certain number of people have given the same answer.

The best part is that nobody has to go around soliciting help from Web users—they’re doing what they would do anyway, but their activities have an effect they don’t fathom.

Getting Sneaky, Part II

You only use 10 percent of your brain, they say—